

MIT Ideator

Neuro-symbolic Ideation Machine

Problem:

- Current Ideation methods lack complexity, diversity or effective representation of reasoning and logic.
- ‘Pregnant Question’ approach creates a new direction for future CogSci research.
 - *Pregnant questions*: Questions about an idea or its implementation that result in more questions, meant to raise critical nuances for the reasoning process.

Related Work:

- <https://arxiv.org/abs/2311.01937>
- <https://dl.acm.org/doi/10.1145/3643562.3672611>
- <https://cci.mit.edu/wp-content/uploads/2024/10/CCI-AccWPpublic2024-09final.pdf>

Ideator Approach:

1. **Intake:** The user gives a brief like “brainstorm outreach ideas for a robotics club,” adds constraints, available resources, and success criteria.
2. **Seed expansion:** The LLM generates 20 to 50 seed ideas using analogies, constraints, and a few creativity patterns. Each idea is added as a node with fields for description, rationale, cost guess, time guess, risks, and prerequisites.
3. **Structure:** The system links ideas to pros, cons, evidence, metrics, and dependencies, forming a knowledge graph of nodes and edges.
4. **Similarity clustering:** Embeddings group near-duplicate or thematically similar ideas into clusters. The system names each cluster and picks a few representatives.
5. **Reasoning and pruning:** Rule checks run on the graph to flag contradictions, missing prerequisites, legal or ethical concerns, and opportunities to combine ideas. Low-quality ideas are parked, promising ones are refined.
6. **Holistic evaluation:** Each idea receives multi-criteria scores (novelty, value, feasibility, cost, time, risk, impact). The system explains each score using graph evidence and simple calculations. The n strongest ideas are selected.

Approach Benefits

- **Mixes breadth with depth:** embeddings and clusters give breadth while rules and questions focus depth where it matters.
- **Keeps humans in control:** *pregnant questions* translate uncertainty into decisions that reshape the search.
- **It is collaborative:** any website visitor can visualize the knowledge graph, add, remove, or edit nodes, contributing to a certain topic and idea.
- The neurosymbolic approach is a **new field of AI** that is meant for cognitive reasoning and structural analysis. By using a node of attributes to represent an idea, it can be evaluated holistically and within the context of other ideas, benefits, or constraints.

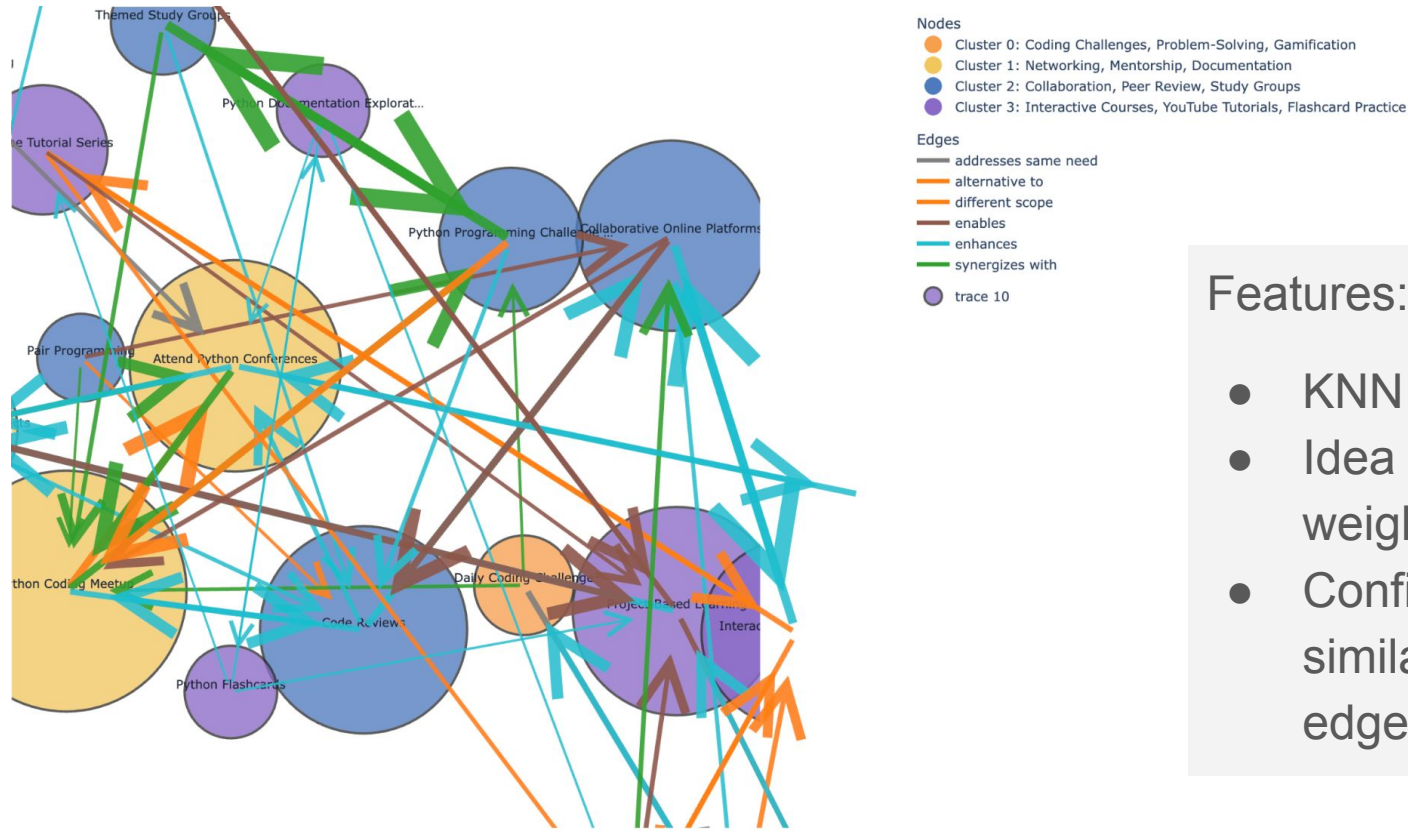
Current Features:

Area	Concrete Change	Why it Helps	How to Implement (Practical Notes)
Embedding Graph	Replace cosine on raw embeddings with Supervised relation head (e.g., XGB/MLP over [cosine, Δ TF-IDF topics, keyword overlap, NLI scores])	Captures richer signals; reduces false “related_to” edges	Build feature matrix per pair; train on small labeled set; fallback to thresholding when low-confidence
Layout	Use UMAP > ForceAtlas2 hybrid positions	Preserves manifold + readable local structure	umap(n_neighbors=10, min_dist=0.2) > fa2(force_atlas2)
Relation Typing	Add NLI-based typing (entailment > refines, contradiction > conflicts)	Makes edges semantically grounded	Use NLI (deBERTa-v3-base-mnli); cache scores
Rule Layer	Typed constraints: prereq, conflicts, mutually_exclusive, budget/time	Prevents invalid plans; improves solver feasibility	Encode in Z3/OR-Tools; maintain rule registry
Scoring	Graph-aware novelty (inverse degree & inverse cluster freq), feasibility (rule penalties), risk (conflict proximity), impact (expert priors)	Makes scores defensible and local to the KG	Compute from KG stats; attach as node attrs

Future Experiments:

Overall/Abl:	Type:	Proves:	How to Execute:
Overall	Stability (Paraphrase & Bootstrap)	Robustness of the full KG & recommendations	Paraphrase titles/desc; rebuild KG; Bootstrap (70%) sampling; rebuild
Overall	Usability & Trust	Cognitive fit and explainability	Short post-task surveys
Overall (Synthetic)	Ground-Truth Stress Test	Accuracy when truth is known	Generate synthetic idea universes w/ known edges & constraints
Ablation	Rules Off	Value of symbolic consistency	Turn off contradiction/missing-prereq checks
Ablation	Solver Off	Value of portfolio optimization	Replace solver with greedy/top-k pick
Ablation	Embeddings > TF-IDF	Value of dense semantics	Swap sentence-embeddings for TF-IDF vectors

Knowledge Graph

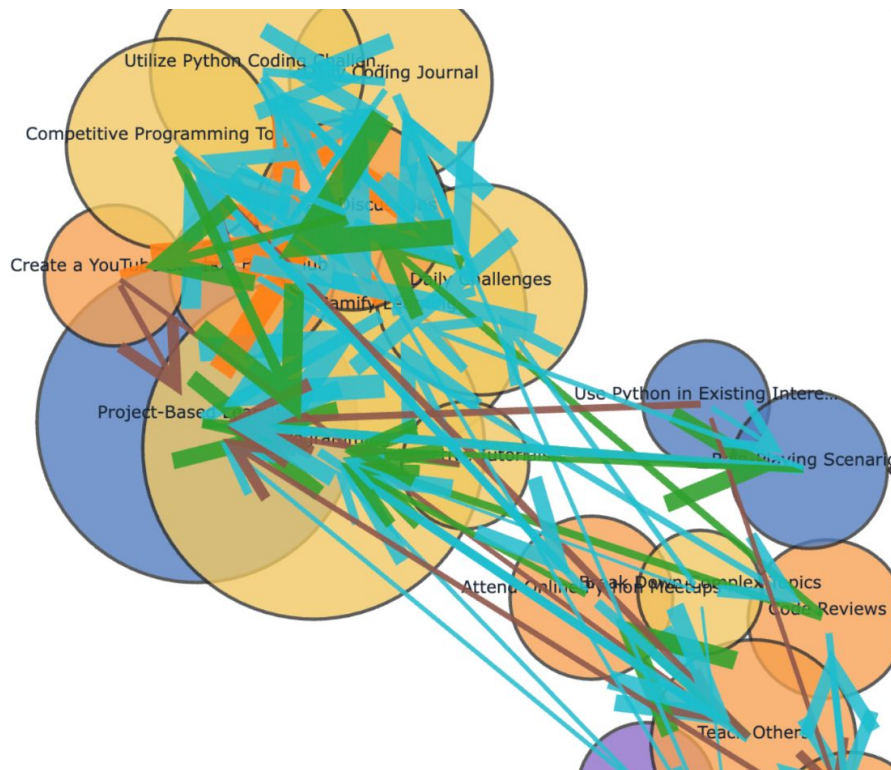


Features:

- KNN Clustering
- Idea relevance & weight determine size
- Confidence and similarity determine edge sizes

Plotly interface: Node clusters with key words & edge key

Hover Tooltip



Role-Playing Scenarios

Simulate real-world scenarios where Python can be used, like data organization, through role-playing exercises.

Details

Constraints: Use Python to solve the simulated problems.

Rationale: Makes learning interactive and relevant through practical application.

Cost: \$0

Time: Varies; several sessions over time

Risks: May not reflect real-world complexities accurately.

Prerequisites: Creativity and an understanding of potential applications of Python.

Questions

1. How might role-playing exercises help participants understand the practical applications of Python in data organization, and what specific scenarios could effectively illustrate common challenges faced in real-world data management?
2. In what ways can varying the complexity of the simulated scenarios enhance participants' problem-solving skills while using Python, and how can these exercises be tailored to different skill levels among learners?
3. What ethical considerations should be taken into account when simulating data organization scenarios, especially regarding privacy and data integrity, and how can role-playing effectively address the importance of these issues in a Python programming context?

Scores

novelty_graph: 5.98

feasibility_graph: 5.00

risk_graph: 5.00

impact_graph: 2.57

combined_graph: 2.89

Hover shows idea/node attributes: heuristic scoring, cost/constraints, pregnant questions

Planner

```
[Diversity filter] Diverse subset of ideas:
```

- Project-Based Learning
- Python Podcast Listening
- Pair Programming
- Python Documentation Exploration
- Daily Coding Challenge
- Python Flashcards
- Python Programming Challenge Groups
- YouTube Tutorial Series
- Mentorship Program
- Code Reviews

```
[Planner] Status: HEURISTIC_SUCCESS
```

```
[Planner] Objective value: 49.27227481598487
```

```
[Planner] Selected ideas:
```

- Python Coding Meetup
- Code Reviews
- Gamified Learning Platforms
- Project-Based Learning
- Daily Coding Challenge
- Attend Python Conferences
- Collaborative Online Platforms
- Interactive Online Course
- Python Programming Challenge Groups
- Work on Open Source Projects
- Interactive Python Books
- YouTube Tutorial Series
- Blog about Learning Journey
- Themed Study Groups
- Create a Portfolio Website
- Mentorship Program
- Pair Programming

- Diversity filter dedupes ideas
- Planner evaluates graph scores holistically, outputs strongest ideas
 - Weighs each sub-score & adds to get total score per idea
 - Outputs n best ideas [user specified]

Current Work/Next Steps:

- Optimizing methods & compute time
- Running ablation studies & baseline experiments